Lab03_FabianoJunior

#1

```r
library(readr)
library(dplyr)
```

```
Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

    filter, lag

Os seguintes objetos são mascarados por 'package:base':

    intersect, setdiff, setequal, union
```

```r
##1
getStats <- function(x, pos) {
  x = x%>%
    filter(!startsWith(DESTINATION_AIRPORT, "1")) %>%
    filter(complete.cases(.)) %>%
  group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atrasos = sum(ARRIVAL_DELAY),
      qtd = n()
    )
  x
}
flights_col = cols_only(DESTINATION_AIRPORT = 'c', ARRIVAL_DELAY = 'd')
flights = read_csv_chunked("flights.csv",
```

```r
          callback = DataFrameCallback$new(getStats),
          chunk_size = 1000000,
          col_types = flights_col)
airlines = read_csv("airlines.csv")
```

```
Rows: 14 Columns: 2

-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
airports = read_csv("airports.csv",
                    col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE))
```

```
Rows: 322 Columns: 5
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (3): IATA_CODE, CITY, STATE
dbl (2): LATITUDE, LONGITUDE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
flights = bind_rows(flights) %>%
  group_by(DESTINATION_AIRPORT)%>%
  summarise(
    soma_atrasos = sum(soma_atrasos),
    qtd = sum(qtd),
    media_atraso = soma_atrasos / qtd
  )
```

#2

```r
##2
### a coluna chave em flights eh DESTINATION_AIRPORT
### a coluna chave em CITY e STATE
### aeroportos que nao tem em airports mas tem em flights
```

```r
ausentes <- flights %>%
  anti_join(
    airports,
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )
### todos aeroportos estao listados

flights <- left_join(
  flights,
  airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)
```

#3

```r
airports %>%
  group_by(STATE) %>%
  summarise(qtd_aeroportos = n()) %>%
  arrange(desc(qtd_aeroportos))
```

```
# A tibble: 54 x 2
   STATE qtd_aeroportos
   <chr>          <int>
 1 TX                24
 2 CA                22
 3 AK                19
 4 FL                17
 5 MI                15
 6 NY                14
 7 CO                10
 8 MN                 8
 9 MT                 8
10 NC                 8
# i 44 more rows
```

#4

```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
library(leaflet)
this = leaflet(flights) %>%
```

```
  addTiles()%>%
  addMarkers(
  lng = ~LONGITUDE,
  lat = ~LATITUDE,
  popup = ~paste(
      "Aeroporto:", DESTINATION_AIRPORT,
      "<br>Atraso médio:", media_atraso
  ))
this
```